Sample-Efficient FunSearch via Behavioral Fingerprinting for Online Bin Packing

CS5491 Artificial Intelligence · Final Project · FunSearch Enhancements

Abstract

We compare a fixed 16-state behavioral filter, S2-N16, with Standard FunSearch and exact-source filtering in three matched live pairs. At 180 generation requests per method, S2-N16 completed 48 full evaluations versus 83 for Exact and 180 for Standard, a 42.2% reduction versus Exact. Its active time was lower in two pairs and higher in one. An audit of 18 skips found two fitness mismatches and seven potential selected-island gains. Development and held-out fitness tied in two pairs. In the third, S2-N16 improved equal-family fitness by using fewer Weibull bins and more OR3 bins.

1. Motivation and background

FunSearch generates heuristic functions with an LLM, evaluates them, and uses an island database of selected programs to construct later prompts [1–3]. In online bin packing, items arrive sequentially and must be placed without seeing future items. The evaluator calls the candidate's priority(item, bins) on feasible residual capacities, selects the first maximum, and opens a new bin if none fits. The aim is to minimize bins used. First Fit and Best Fit provide packing baselines.

Different source programs can make the same decisions. We measure whether a short decision trace saves evaluation time and how screening affects search feedback and packing quality.

2. Fixed probes and processing policy

S2 collects feasible states along First Fit, Best Fit, Worst Fit and seeded random Development prefixes. It stratifies by family, four item-size bands cut at ¼, ½ and ¾ of capacity, and two layouts with an empty bin and either one occupied feasible bin or at least two distinct occupied residual capacities. Selection favors available exact fits and tied capacities, uses hash order to break ties, and redistributes empty strata. Rounds of 16 distinct states form fixed nested 16/64/128 banks. A separate Development-side construction provides 256 validation states [E1].

For state s = (x, r, I), x is the item, r the feasible capacities and I their original bin indices. The fingerprint records the selected index on each fixed probe using the evaluator's tie rule.

$$d_h(s) = I[\text{first arg max}_j\ p_h(x, r)_j], \quad \Phi_N(h) = (d_h(s_1), \ldots, d_h(s_N)).$$

After static checks, Standard fully evaluates candidates. Exact skips matching normalized source. S2-N16 applies Exact, then reuses a recorded behavioral skip for the same source (C1). Remaining candidates run the probes. A vector match marks a suspected duplicate of an evaluated representative. A miss proceeds to full evaluation, as does a recoverable probe failure (F1). Only complete valid results register. A matching vector establishes agreement on these probes; it cannot certify equivalence or equal full-stream fitness.

3. Fixed-pool replay and live controls

Replay. Every replay arm received the same ordered 16-request candidate stream, with no fresh generation. In the state-policy replay, S2-16/64/128 each completed three full evaluations per block versus eight for Exact, retained the same best fitness, and separated the known #10/#8 false match that S0 missed. N16 was the smallest tested S2 bank that achieved those observed outcomes. Larger banks supplied no additional discrimination on this pool. The earlier separate-worker replay took longer than Exact despite saving evaluations. The three sizes were compared in replay; only N16 entered the live study [E1].

Live search. Three matched pairs used 60 requests per method and pair (540 main requests). Methods started from matched checkpoints with independent databases, paired decoding seeds and rotating serial order. Skips and registrations could change later prompts and candidates. Full evaluation used 12 OR3 and three Weibull Development instances [1, 4]. Fitness is higher-is-better, with 50% weight per family.

$$F(h) = -\tfrac{1}{2}\,[\text{mean bins}_{OR3}(h) + \text{mean bins}_{Weibull}(h)].$$

The fixed generator was local Ollama Qwen3.8 27B Q4_K_M on an NVIDIA RTX 5090, with temperature 0.2, top-p 0.9, 512 output tokens, 32,768-token context and reasoning disabled. It replaced the proposed Qwen2.5-Coder-7B-Instruct. Recorded usage shows no DeepSeek fallback and $0 in external API charges. Hardware and electricity costs were unmetered.

4. Efficiency and timing

Method	Requests	Full evals	Containers	Generation (s)	Evaluation (s)
Standard	180	180	180	978.605	1,015.787
Exact	180	83	83	986.405	763.814
S2-N16	180	48	99	925.612	748.887

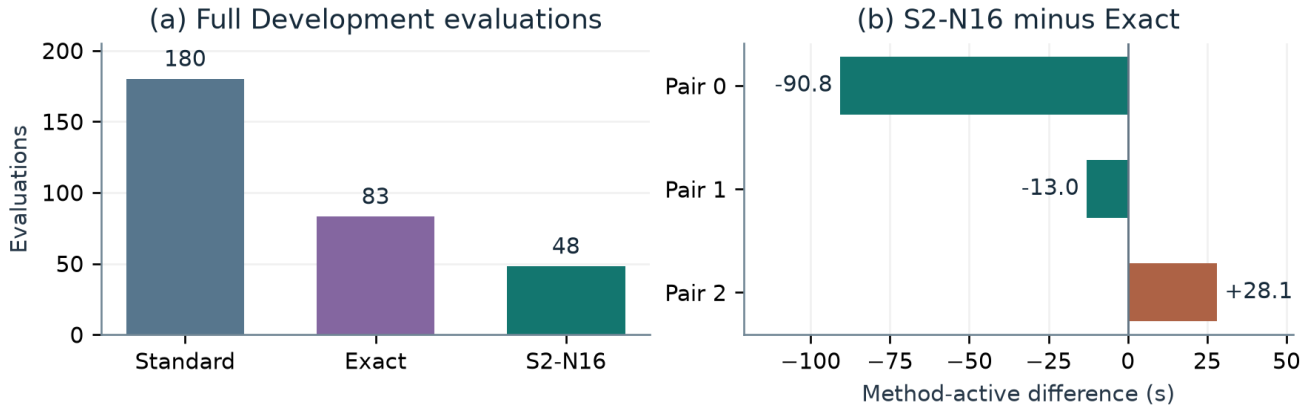


Figure 1. Full Development evaluations and paired method-active differences versus Exact.

S2-N16 saved 42.2% versus Exact and 73.3% versus Standard. Against Exact, its method-active differences were −90.801, −13.049 and +28.131 s. Pair 2 saved ten full evaluations while using seven extra containers. Generation plus evaluation time includes method-owned readiness, filtering, worker and registration costs. Human approval, shared orchestration, downtime and independent audit time are excluded. Evaluation savings did not consistently translate into lower active time.

5. Development packing quality

Pair	Method	Final fitness	OR3 mean bins	Weibull mean bins
0	All methods	−1,142.542	212.417	2,072.667
1	All methods	−1,142.542	212.417	2,072.667
2	Standard and Exact	−1,142.542	212.417	2,072.667
2	S2-N16	−1,119.625	225.583	2,013.667

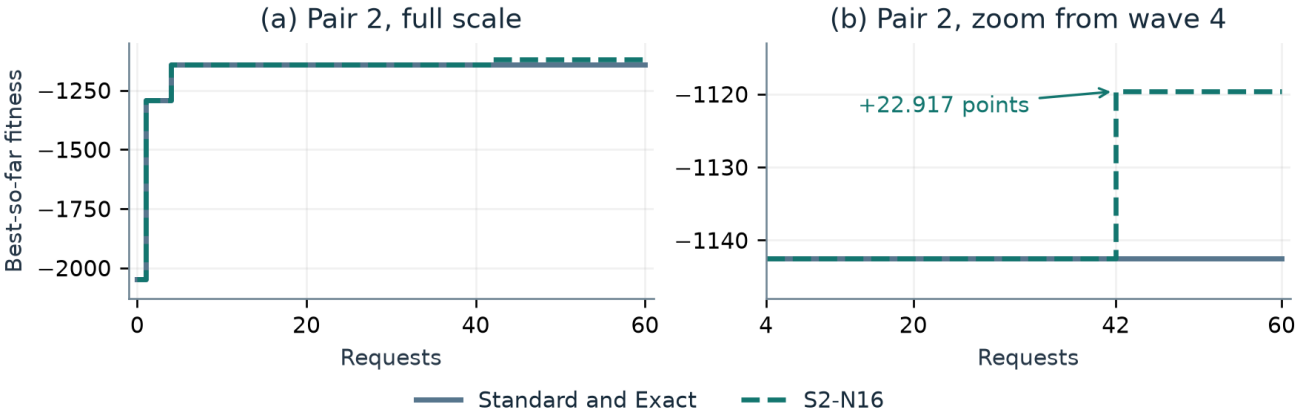


Figure 2. Pair 2 Development trajectory at full scale and zoomed after wave 4. The jump at wave 42 gives S2-N16 a 22.917-point final advantage over both controls.

Pairs 0 and 1 tie. In Pair 2, S2-N16 uses 13.167 more OR3 bins and 59.000 fewer Weibull bins per instance than both controls. Its equal-family fitness improves by 22.917 points. This is the observed live outcome. Which individual search decision produced the gain is unresolved.

6. Independent audit of skips

A frozen hash rank selected six direct behavioral or C1 skip events per pair from 68 eligible events. The 18 sampled events covered 24 distinct sources across candidates and representatives. Each received one independent full Development evaluation and a vector of 256 separate validation decisions. Opportunity margins compare the remeasured candidate with the historical pre-skip threshold, using a tolerance of 1e-9.

Pair	Events	Fitness mismatches	Vector-mismatch events	Island opportunities	Above global best
0	6	0	0	4	0
1	6	1	0	2	0
2	6	1	1	1	0
Total	18	2	1	7	0

Pair 1 #129 scored 1.208 points above #122 despite agreeing on all 256 decisions. Pair 2 #166 scored 6.917 below #139 and differed at two validation positions. In six of the seven potential island improvements, candidate and representative fitness tied. The selected island's historical threshold can still be below that fitness. The seven margins measure local opportunities. Alternative registrations and downstream trajectories were unobserved. The deterministic sample and repeated sources do not estimate population false-match rates [E2].

7. Held-out packing quality

The held-out addendum evaluated the final Development incumbent of each run, using lowest island ID to break fitness ties. Nine provenance rows map to eight measured source/body identities because Pair 0 Exact and S2-N16 share one source. The held-out split contains four 500-item OR3 instances and one 5,000-item Weibull instance.

Pair	Method	OR3 mean bins	Weibull bins	Fitness
0	All methods	209.75	2,058	−1,133.875
1	All methods	209.75	2,058	−1,133.875
2	Standard and Exact	209.75	2,058	−1,133.875
2	S2-N16	224.00	2,000	−1,112.000

Pairs 0 and 1 tie. Pair 2 S2-N16 uses 14.25 more OR3 bins per instance and 58 fewer Weibull bins. Its equal-family fitness is 21.875 points higher. The direction repeats the Development trade-off. The 50/50 weighting gives one Weibull instance the same aggregate influence as the four OR3 instances together.

8. Research accounting and limitations

The proposal estimated approximately 570 requests (30 pilot plus 540 main). The known pre-main ledger reached 51: 30 earlier logged requests, three rejected V3 outputs, and 18 calls in the revised-prompt V4 pilot. These include documented pre-main troubleshooting and prompt revision. Earlier connectivity usage is unresolved. The 540 matched main requests therefore give 591 classified logged requests across those scopes, with the exact lifetime total unknown. Readiness preloads and infrastructure repairs are accounted separately.

Independent audit used 24 full and 24 validation workers, with zero model requests. The held-out ledger retains one charged startup failure, a zero-charge guard stop, and eight completed evaluations. A prior inert diagnostic container is separate. The audit-event table, resource receipts and nine held-out rows are indexed in the repository supplement [E2].

The OR3 held-out split was designated retrospectively after historical whole-OR3 baseline evaluation and legacy reference-data initialization. Five instances provide limited evidence for generalization. Three matched live comparisons on one host also provide limited evidence about timing and search quality. Divergent prompts prevent attributing the Pair 2 result to a specific skip. Finite probes and the deterministic audit quantify observed discrepancies on the sampled inputs.

9. Conclusion

At a matched generation budget, S2-N16 saved 35 full Development evaluations versus Exact. Active-time differences varied across pairs. The audit found fitness mismatches and possible island-level feedback losses. The favorable Pair 2 aggregate reflects a trade-off between instance families. Behavioral fingerprints are a practical screening heuristic whose value depends on probe coverage, search history and evaluation cost.

References

[1] B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli and A. Fawzi. “Mathematical discoveries from program search with large language models.” *Nature*, 625(7995), 468–475, 2024. doi:10.1038/s41586-023-06924-6.

[2] Google DeepMind. FunSearch official source repository, online bin-packing evaluator and program search implementation. Apache-2.0 software license.

[3] RayZhhh. Course-linked FunSearch bin-packing implementation, commit 4b8b4e7d43287998500983d709dfa44be58fca91. The repository NOTICE records the local source attribution.

[4] J. E. Beasley. OR-Library: one-dimensional bin packing, binpack3 uniform 500-item instances contributed by E. Falkenauer. Weibull benchmark methodology follows [1, 2].

Project evidence (internal records and repository supplements)

[E1] CS5491 State Policy v2, docs/STATE_POLICY_V2_RESULTS.md, phase3/state_policy_v2/states.py and results/state_policy_v2/. Frozen state construction, replay, and per-size measurements.

[E2] CS5491 completed study, independent audit and held-out archive. paper/SUPPLEMENT.md indexes the extended report and results/main_study/ tables. The latter contains receipt-derived observations and provenance digests. REPRODUCIBILITY.md specifies offline checks and runtime limits.